The Discriminant

A Road into Quadratic and Cubic Reciprocity

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Abstract

One familiar object, the discriminant, is followed through two levels of arithmetic. For a quadratic it detects solvability modulo a prime and leads to the Legendre symbol and quadratic reciprocity. For a pure cubic $t^3 - a$, the value $-27a^2$ singles out $\mathbb{Q}(\sqrt{-3})$, while the three roots force the cube roots of unity into the same field. Its ring of integers is the Eisenstein ring $\mathbb{Z}[\omega]$, the natural home of the cubic residue symbol and cubic reciprocity. Hand calculations precede the code, and two colour-coded matrices make the contrasting symmetry laws visible.

1 The quadratic touchstone

For

$$ax^2 + bx + c = 0$$

the discriminant is

$$D = b^2 - 4ac.$$

Over $\mathbb{R}$, its sign distinguishes two real roots, a repeated root, and a complex-conjugate pair. We now ask the same equation to hold modulo an odd prime p. The condition

$$p \nmid a$$

simply means that p does not divide the leading coefficient a. It ensures that $2a$ has a multiplicative inverse modulo p, so division by $2a$ remains possible in modular arithmetic. The complete-square step comes from the ordinary algebraic identity

$$\begin{aligned}(2ax + b)^2 &= 4a^2x^2 + 4abx + b^2 \\ &= 4a(ax^2 + bx + c) + (b^2 - 4ac) \\ &= 4a(ax^2 + bx + c) + D.\end{aligned}$$

If x is a solution of

$$ax^2 + bx + c \equiv 0 \pmod{p},$$

then the first term on the final line vanishes modulo p, leaving

$$(2ax + b)^2 \equiv D \pmod{p}.$$

Thus a solution of the quadratic produces a square root of D modulo p. The reasoning also runs backwards. If $y^2 \equiv D \pmod{p}$, then

$$2ax + b \equiv y \pmod{p}$$

can be solved for x because $2a$ is invertible:

$$x \equiv (y-b)(2a)^{-1} \pmod{p}.$$

For this value of x, the identity above gives $4a(ax^2+bx+c) \equiv y^2 - D \equiv 0 \pmod{p}$; since $4a$ is invertible, the quadratic itself is congruent to zero. The quadratic therefore has a solution modulo p exactly when D is a square modulo p, including $D \equiv 0 \pmod{p}$. Mathematicians denote the arithmetic of the p possible remainders by $\mathbb{F}_p$; saying that the quadratic has a root in $\mathbb{F}_p$ means precisely that it has a solution modulo p.

Definition 1 (Legendre symbol)**.** For an odd prime p,

$$\left(\frac{u}{p}\right) = \begin{cases} 0, & p \mid u, \\ +1, & u \text{ is a nonzero square modulo } p, \\ -1, & u \text{ is not a square modulo } p. \end{cases}$$

Thus $\left(\frac{u}{p}\right)$ records whether $x^2 \equiv u \pmod{p}$ is solvable.

1.1 A first hand calculation

Take $p = 7$. Squaring the nonzero residues gives

x	1	2	3	4	5	6
x^2 mod 7	1	4	2	2	4	1

so the nonzero squares modulo 7 are $1, 2, 4$. Hence

$$\left(\frac{2}{7}\right) = +1 \quad \text{because} \quad 3^2 \equiv 2 \pmod{7}, \qquad \left(\frac{3}{7}\right) = -1.$$

Accordingly, $x^2 - 2 \equiv 0 \pmod{7}$ has the two solutions $x \equiv 3, 4$, while $x^2 - 3 \equiv 0 \pmod{7}$ has none.

1.2 Euler's criterion: the hand table becomes an algorithm

The table suggests a shortcut. Instead of squaring every number, raise the number being tested to the power $(p-1)/2$. For an odd prime p that does not divide u, the answer is always 1 or -1 modulo p:

$$u^{(p-1)/2} \equiv \left(\frac{u}{p}\right) \pmod{p}. \tag{1}$$

For example, modulo 7,

$$2^3 \equiv 1 \pmod{7}, \qquad 3^3 \equiv -1 \pmod{7}.$$

This recovers the result of the table: 2 is a square modulo 7, while 3 is not.

Here is why the shortcut works. Fermat's little theorem gives

$$u^{p-1} \equiv 1 \pmod{p}.$$

Consequently $u^{(p-1)/2}$ is a number whose square is 1. Modulo a prime, the only possibilities are 1 and -1, since

$$x^2 - 1 = (x-1)(x+1).$$

If u is a square, say $u \equiv r^2 \pmod{p}$, then

$$u^{(p-1)/2} \equiv r^{p-1} \equiv 1 \pmod{p}$$

by Fermat's theorem again.

There are exactly $(p-1)/2$ nonzero squares modulo p: the numbers r and $p-r$ have the same square, so the $p-1$ nonzero numbers pair up. Meanwhile, the polynomial

$$x^{(p-1)/2} - 1$$

can have at most $(p-1)/2$ roots modulo p. The known squares already supply all of them. Every remaining number must therefore give -1. This result is Euler's criterion, the hinge between the hand table and the later code.

1.3 Quadratic reciprocity as an exchange law

For distinct odd primes p and q, quadratic reciprocity states

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}. \tag{2}$$

The two symbols agree unless $p \equiv q \equiv 3 \pmod 4$, when they have opposite signs.

For $p = 3$ and $q = 7$,

$$\left(\frac{3}{7}\right) = -1, \qquad \left(\frac{7}{3}\right) = \left(\frac{1}{3}\right) = +1.$$

For $p = 3$ and $q = 5$,

$$\left(\frac{3}{5}\right) = -1, \qquad \left(\frac{5}{3}\right) = \left(\frac{2}{3}\right) = -1.$$

The congruence classes modulo 4 determine whether the exchanged symbols agree; they do not determine either symbol separately.

1.4 A brief view of splitting

If d is a nonsquare integer, adjoining $\sqrt{d}$ to $\mathbb{Q}$ produces the quadratic field $\mathbb{Q}(\sqrt{d})$. For an odd prime p away from the field discriminant, the value $\left(\frac{d}{p}\right)$ records whether p splits into two prime factors or remains prime in the ring of integers of the field. The value $+1$ means split and -1 means inert. This is the number-field version of the elementary question whether $x^2 \equiv d \pmod p$ has a solution. We shall need only the special field $\mathbb{Q}(\sqrt{-3})$ later.

Listing 1: Euler's criterion computes the Legendre symbol.

```python
def legendre(a: int, p: int) -> int:
    """Return the Legendre symbol (a/p) for an odd prime p."""
    a %= p
    if a == 0:
        return 0
    value = pow(a, (p - 1) // 2, p)
    return 1 if value == 1 else -1

residues_13 = sorted({k * k % 13 for k in range(1, 13)})
print(residues_13)          # [1, 3, 4, 9, 10, 12]
print(legendre(3, 13))      # +1, since 4^2 = 3 (mod 13)
print(legendre(3, 5))       # -1
```

2 The cubic discriminant

Cardano's formula for the general depressed cubic

$$t^3 + At + B = 0.$$

Its discriminant is

$$\Delta = -4A^3 - 27B^2,$$

and Cardano's expression contains

$$\sqrt{\left(\frac{B}{2}\right)^2 + \left(\frac{A}{3}\right)^3} = \sqrt{-\frac{\Delta}{108}}.$$

Over $\mathbb{R}$, the sign of the quantity under this square root distinguishes one real root, a repeated root, and the casus irreducibilis of three distinct real roots reached through complex intermediate quantities. This locates the cubic discriminant inside the familiar formula; the formula itself will play no further role.

Boundary marker 1 (Why we now restrict the question)**.** For a general cubic, knowing whether Δ is a square modulo p gives useful information about its factorisation, although it does not by itself distinguish complete splitting from irreducibility. Completing that general story leads beyond elementary cubic residues and towards Galois and class field theory. From this point onward we study the pure cubic $t^3 - a$. Here the cubic residue symbol answers exactly the question we ask. Its discriminant is $-27a^2 \leq 0$, so a pure cubic never enters the casus irreducibilis. The route through the general cubic's triple-angle substitution therefore closes here; the pure cubic reaches the cube roots of unity by a shorter and more direct road, through its three complex roots.

2.1 The pure cubic forces the field $\mathbb{Q}(\sqrt{-3})$

For

$$f(t) = t^3 - a$$

the discriminant is

$$\Delta = -27a^2.$$

When $a \neq 0$,

$$\mathbb{Q}(\sqrt{\Delta}) = \mathbb{Q}(\sqrt{-3}).$$

The same field appears directly from the roots. If $r^3 = a$ and

$$\omega = e^{2\pi i/3}, \qquad \omega^2 + \omega + 1 = 0,$$

then

$$t^3 - a = (t - r)(t - \omega r)(t - \omega^2 r).$$

One cube root brings the others with it only after ω has been adjoined. The ring of integers of $\mathbb{Q}(\sqrt{-3})$ is

$$\mathbb{Z}[\omega] = \{x + y\omega \,:\, x, y \in \mathbb{Z}\},$$

the ring of Eisenstein integers. Its norm is

$$N(x + y\omega) = (x + y\omega)(x + y\omega^2) = x^2 - xy + y^2.$$

Thus the discriminant and the roots of the pure cubic independently lead to the same arithmetic setting.

2.2 Cubes modulo an ordinary prime

Modulo 7,

x	1	2	3	4	5	6
x^3 mod 7	1	1	6	1	6	6

so the nonzero cubes are 1 and 6. The map $x \mapsto x^3$ on $\mathbb{F}_p^\times$ has a kernel of size 3 when $p \equiv 1 \pmod 3$. Consequently its image contains $(p-1)/3$ elements, and the nonzero elements fall into three cubic classes.

If instead $p \equiv 2 \pmod 3$, then 3 is coprime to $p - 1$. Cubing is a bijection on $\mathbb{F}_p^\times$, so every nonzero element is a cube. The interesting three-way distinction therefore occurs at primes $p \equiv 1 \pmod 3$.

3 The cubic residue symbol in $\mathbb{Z}[\omega]$

A rational prime $p \equiv 1 \pmod 3$ splits in the Eisenstein integers:

$$p = \pi\bar{\pi}, \qquad N\pi = p.$$

Equivalently, for a rational prime $p \neq 3$,

$$p \equiv 1 \pmod 3 \quad \iff \quad p = a^2 - ab + b^2 \quad \text{for some } a, b \in \mathbb{Z}. \tag{3}$$

This classical representation theorem explains why the norm search in Listing 2 succeeds; see Cox [1] for the wider theory of primes represented by quadratic forms.

There are two conjugate primes above p. This choice will matter to the label ω or ω^2, so we shall make it explicit.

Definition 2 (Primary and canonical primary primes)**.** Let $\pi = a + b\omega$ with $3 \nmid N\pi$. It is primary when

$$\pi \equiv 2 \pmod 3, \qquad \text{equivalently} \qquad a \equiv 2 \pmod 3, \quad b \equiv 0 \pmod 3.$$

Among the six associates $\pi, -\pi, \omega\pi, -\omega\pi, \omega^2\pi, -\omega^2\pi$, exactly one is primary. If $\pi = a + b\omega$ is primary, then

$$\bar{\pi} = (a - b) - b\omega$$

is primary as well: $a - b \equiv 2 \pmod 3$ and $-b \equiv 0 \pmod 3$. Thus the two conjugate primes above every $p \equiv 1 \pmod 3$ have primary representatives with opposite signs of the ω-coefficient. In this paper the canonical primary prime above p is the one written $a + b\omega$ with $b > 0$.

For example,

$$7 = (2 + 3\omega)(2 + 3\omega^2).$$

In the $a + b\omega$ notation the two primary primes are

$$2 + 3\omega \qquad \text{and} \qquad -1 - 3\omega.$$

Our $b > 0$ convention selects $2 + 3\omega$.

Definition 3 (Cubic residue symbol)**.** Let π be an Eisenstein prime with $N\pi \neq 3$, and suppose $\pi \nmid \alpha$. The cubic residue symbol is the unique element of $\{1, \omega, \omega^2\}$ satisfying

$$\alpha^{(N\pi - 1)/3} \equiv \left(\frac{\alpha}{\pi}\right)_3 \pmod \pi.$$

The symbol equals 1 precisely when α is a cube modulo π.

This is the cubic analogue of Euler's criterion. The values ω and ω^2 distinguish the two non-cube classes rather than merely reporting "not a cube."

3.1 A cubic symbol by hand

Let $\pi = 2 + 3\omega$, the canonical primary prime above 7. In the quotient $\mathbb{Z}[\omega]/(\pi) \cong \mathbb{F}_7$,

$$2 + 3\omega \equiv 0 \implies \omega \equiv -2 \cdot 3^{-1} \equiv 4 \pmod 7.$$

Thus $\omega^2 \equiv 2 \pmod 7$. For $\alpha = 2$,

$$2^{(N\pi - 1)/3} = 2^2 \equiv 4 \equiv \omega \pmod \pi,$$

so

$$\boxed{\left(\frac{2}{2 + 3\omega}\right)_3 = \omega.}$$

Likewise $3^2 \equiv 2 \equiv \omega^2 \pmod 7$, giving

$$\left(\frac{3}{2 + 3\omega}\right)_3 = \omega^2.$$

3.2 Cubic reciprocity in numbers

For primary Eisenstein primes π and λ with distinct norms, neither of norm 3, Eisenstein's cubic reciprocity law states

$$\left(\frac{\pi}{\lambda}\right)_3 = \left(\frac{\lambda}{\pi}\right)_3. \tag{4}$$

(See Ireland and Rosen [2, Chapter 9].) Take the canonical choices

$$\pi = 2 + 3\omega, \quad N\pi = 7, \qquad \lambda = -1 + 3\omega, \quad N\lambda = 13.$$

Modulo π, $\omega \equiv 4 \pmod 7$, so

$$\lambda \equiv -1 + 3(4) \equiv 4 \pmod 7, \qquad \lambda^2 \equiv 2 \equiv \omega^2 \pmod \pi.$$

Hence $\left(\frac{\lambda}{\pi}\right)_3 = \omega^2$.

Modulo λ, $\omega \equiv 3^{-1} \equiv 9 \pmod{13}$. Therefore

$$\pi \equiv 2 + 3(9) \equiv 3 \pmod{13}, \qquad \pi^4 \equiv 3 \equiv \omega^2 \pmod \lambda.$$

Thus

$$\boxed{\left(\frac{\lambda}{\pi}\right)_3 = \left(\frac{\pi}{\lambda}\right)_3 = \omega^2.}$$

This equality is the contrast that the matrices below are built to make visible.

4 Computing without hiding the conventions

If $\pi = a + b\omega$ has norm p, then in $\mathbb{Z}[\omega]/(\pi) \cong \mathbb{F}_p$,

$$\omega \equiv -ab^{-1} \pmod p =: r_\pi.$$

The reduction map is therefore

$$\varphi_\pi(x + y\omega) = x + yr_\pi \pmod p.$$

The code below returns the exponent 0, 1, or 2 in ω^k, and names the function accordingly. It also uses an explicit extended Euclidean algorithm, so it does not depend on the modular-inverse syntax introduced in Python 3.8.

Listing 2: Canonical primary primes and cubic-symbol exponents.

```
from math import isqrt

def inverse_mod(a: int, m: int) -> int:
    """Return a^(-1) modulo m using the extended Euclidean algorithm."""
    old_r, r = m, a % m
    old_s, s = 0, 1       # coefficients of a in old_r and r
    while r:
        q = old_r // r
        old_r, r = r, old_r - q * r
        old_s, s = s, old_s - q * s
    if old_r != 1:
        raise ValueError("inverse does not exist")
    return old_s % m

def associates(a: int, b: int):
    """The six unit multiples of a+b*w, as coefficient pairs."""
    return [(a, b), (-b, a - b), (b - a, -a),
            (-a, -b), (b, b - a), (a - b, a)]

```

```python
def primary_associate(a: int, b: int):
    """Return the unique primary element among the six associates."""
    for x, y in associates(a, b):
        if x % 3 == 2 and y % 3 == 0:
            return x, y
    raise ValueError("no primary associate")

def norm_representation(p: int):
    """Find one a,b with a^2-a*b+b^2=p using a one-dimensional search."""
    for b in range(1, isqrt(4 * p // 3) + 2):
        disc = 4 * p - 3 * b * b
        if disc < 0:
            break
        root = isqrt(disc)
        if root * root == disc and (b + root) % 2 == 0:
            return (b + root) // 2, b
    raise ValueError("p has no Eisenstein norm representation")

def conjugate(z):
    """Conjugate a+b*w: a+b*w^2=(a-b)-b*w."""
    a, b = z
    return a - b, -b

def canonical_primary(p: int):
    """Choose the primary prime a+b*w above p with b>0."""
    a, b = primary_associate(*norm_representation(p))
    if b < 0:
        a, b = conjugate((a, b))
    return a, b

def cubic_symbol_exponent(alpha, pi, p: int) -> int:
    """Return k in {0,1,2} for (alpha/pi)_3=w^k."""
    a, b = pi
    r = (-a * inverse_mod(b, p)) % p        # image of w modulo pi
    x, y = alpha
    reduced = (x + y * r) % p
    if reduced == 0:
        raise ValueError("symbol is zero because pi divides alpha")
    value = pow(reduced, (p - 1) // 3, p)
    for k, target in enumerate((1, r, r * r % p)):
        if value == target:
            return k
    raise ValueError("unexpected cubic-character value")
```

5 Two reciprocity matrices

Both matrices index rows and columns by primes. The diagonal is grey because the corresponding symbol is zero rather than one of the nonzero character values. Figure 1 uses a colour-blind-safe blue/orange pair. Its asymmetry records the correction term in quadratic reciprocity. Figure 2 uses three distinguishable colours and is symmetric.

Listing 3: Checking reciprocity and the conjugate convention.

```python
def primes_below(n: int):
    """A small sieve, sufficient for constructing the displayed matrix."""
    sieve = [True] * n
    sieve[0:2] = [False, False]
```

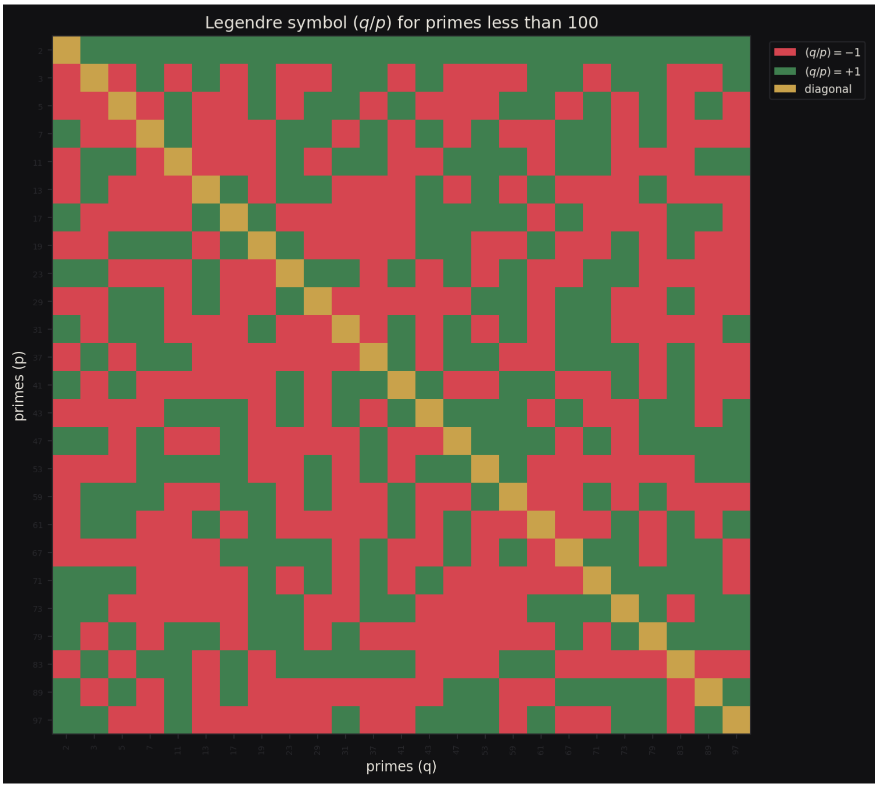

Figure 1: Legendre symbols (q/p) for distinct odd primes below 100. The value in a cell depends on the primes themselves, not merely on their congruence classes modulo 4. Those congruence classes determine the disagreement pattern between the transposed cells (q/p) and (p/q). Blue and orange represent $+1$ and -1; grey marks the diagonal.

```
    for k in range(2, isqrt(n - 1) + 1):
        if sieve[k]:
            sieve[k * k:n:k] = [False] * (((n - 1 - k * k) // k) + 1)
    return [k for k in range(n) if sieve[k]]

primes = [p for p in primes_below(100) if p % 3 == 1]
chosen = {p: canonical_primary(p) for p in primes}

def matrix_is_symmetric(data):
    prime_list = sorted(data)
    return all(
        cubic_symbol_exponent(data[q], data[p], p)
        == cubic_symbol_exponent(data[p], data[q], q)
        for p in prime_list for q in prime_list if p != q
    )

conjugates = {p: conjugate(chosen[p]) for p in primes}
print(matrix_is_symmetric(chosen))        # True
print(matrix_is_symmetric(conjugates))    # True

# Simultaneous conjugation swaps exponents 1 and 2 and fixes 0.
labels_conjugate = all(
    cubic_symbol_exponent(conjugates[q], conjugates[p], p)
    == (-cubic_symbol_exponent(chosen[q], chosen[p], p)) % 3
    for p in primes for q in primes if p != q
)
```

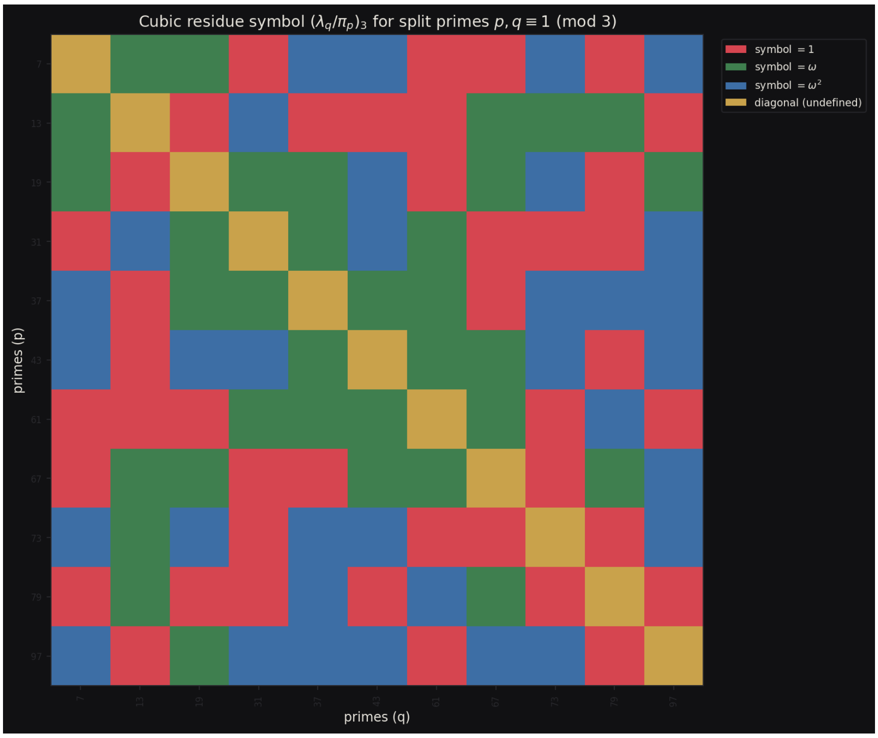

Figure 2: Cubic residue symbols $(\lambda_q/\pi_p)_3$ for primes $p, q \equiv 1 \pmod 3$ below 100. For every rational prime, π_p is the canonical primary prime $a + b\omega$ with $b > 0$. Choosing all conjugate primes instead conjugates the labels $\omega \leftrightarrow \omega^2$ while preserving the symmetry. Grey marks the diagonal.

```
print(labels_conjugate)                    # True
```

5.1 Why one law has a twist and the other does not

Quadratic reciprocity contains the explicit correction factor in (2). It forces disagreement precisely when both primes are 3 modulo 4. Cubic reciprocity for primary Eisenstein primes has the cleaner form (4).

The primary convention is essential, although the absence of a correction term should not be explained merely by counting units. The Eisenstein field already contains the cube roots of unity, and the full explanation belongs to the local and global arithmetic of that field. In more advanced language it is expressed through cubic Hilbert symbols and their product formula at the ramified places. Developing that theory would take us well beyond the present route. The honest conclusion here is narrower: after primary normalisation, Eisenstein's cubic reciprocity law is symmetric, while Gauss's quadratic law retains its sign. Nor should one infer that reciprocity laws become cleaner as the exponent rises. Biquadratic reciprocity in $\mathbb{Z}[i]$, for example, retains supplementary correction factors after its analogous primary normalisation; see Lemmermeyer [3].

6 Exercises

1. List the nonzero squares modulo 11 and use your list to evaluate $\left(\frac{2}{11}\right)$, $\left(\frac{3}{11}\right)$ and $\left(\frac{5}{11}\right)$.

2. Use Euler's criterion to calculate $\left(\frac{3}{13}\right)$. Then exhibit a square root of 3 modulo 13.

3. Without listing all squares, use quadratic reciprocity to compare $\left(\frac{11}{19}\right)$ and $\left(\frac{19}{11}\right)$.

4. Explain why the homomorphism $x \mapsto x^3$ on $\mathbb{F}_p^\times$ has exactly $(p-1)/3$ values when $p \equiv 1 \pmod 3$, and why it is a bijection when $p \equiv 2 \pmod 3$.

5. Verify that $13 = N(-1 + 3\omega)$, show that $-1 + 3\omega$ is primary, and identify its conjugate primary prime.
6. Working modulo $\pi = 2 + 3\omega$, evaluate $\left(\frac{4}{\pi}\right)_3$ and $\left(\frac{5}{\pi}\right)_3$ by hand.
7. Replace $\pi = 2 + 3\omega$ by its conjugate $-1 - 3\omega$ in the preceding exercise. Explain why the answers are complex conjugates of the original symbol values. Hint: conjugate the defining congruence for the cubic symbol.

Terse answers and checks

1. The nonzero squares are $1, 3, 4, 5, 9$. Hence $\left(\frac{2}{11}\right) = -1$, $\left(\frac{3}{11}\right) = +1$ and $\left(\frac{5}{11}\right) = +1$.
2. $3^6 \equiv 1 \pmod{13}$, so $\left(\frac{3}{13}\right) = +1$; the square roots are 4 and 9.
3. Since $11 \equiv 19 \equiv 3 \pmod 4$, the two symbols have opposite signs.
4. The kernel has size $\gcd(3, p-1)$. Apply the homomorphism theorem to the group of size $p - 1$.
5. $N(-1 + 3\omega) = 1 + 3 + 9 = 13$. Its coefficients are 2 and 0 modulo 3, and its conjugate is $-4 - 3\omega$.
6. With $\omega \equiv 4 \pmod 7$, one finds $4^2 \equiv 2 \equiv \omega^2$ and $5^2 \equiv 4 \equiv \omega$. The symbols are therefore ω^2 and ω, respectively.
7. Conjugation fixes rational integers and exchanges $\omega \leftrightarrow \omega^2$, so it conjugates each symbol value.

7 Closing perspective

The appearance of -3 is now structural rather than decorative. For a pure cubic $t^3 - a$, the discriminant is $-27a^2$, so its quadratic subfield is $\mathbb{Q}(\sqrt{-3})$. Factoring the same polynomial requires the three cube roots $r, \omega r, \omega^2 r$, which again introduces $\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$. Its ring of integers $\mathbb{Z}[\omega]$ is therefore the natural home of cubic residuosity. The quadratic discriminant and the cubic roots have pointed to the same field from two directions.

References

[1] David A. Cox, Primes of the Form $x^2 + ny^2$: Fermat, Class Field Theory, and Complex Multiplication, 2nd ed., Wiley, 2013.

[2] Kenneth Ireland and Michael Rosen, A Classical Introduction to Modern Number Theory, 2nd ed., Graduate Texts in Mathematics 84, Springer, 1990; see Chapter 9.

[3] Franz Lemmermeyer, Reciprocity Laws: From Euler to Eisenstein, Springer Monographs in Mathematics, Springer, 2000.

[4] John Stillwell, Elements of Number Theory, Springer, 2003.